

● HARD

● INFORMATION AND IDEAS

● ANSWER KEY

Central Ideas and Details

03

10 answers · Rationale-rich · Teach from doc

Q1**C**

- C. The root systems of fynbos plants allow the plants to take in so many soil nutrients that forest plants are prevented from flourishing in the fynbos.

Choice C is the best answer because it presents a conclusion about the role of fynbos roots in maintaining the border with the Afro-temperate forest that can be reasonably inferred from the text. The text states that plants in the fynbos have "uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil." The text then describes an experiment where forest tree seedlings grown in plots isolated from fynbos roots grew at a rate five times greater than seedlings in areas near fynbos roots. This strongly suggests that fynbos roots are depleting soil nutrients to such an extent that forest plants can't get enough nutrients from that soil to flourish, thereby hindering the formation of any significant "transition space" between the habitats and helping keep the border between them sharp.

Choice A is incorrect because there is no evidence in the text that fynbos roots physically damage the root systems of forest plants. Instead, the text explicitly describes fynbos plants as having "uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil." This description establishes that fynbos roots are specialized for efficient nutrient absorption. The experiment then confirms this function of the roots by showing that forest seedlings "isolated from the roots of fynbos plants exhibited a growth rate five times greater" than those near fynbos roots. This result demonstrates that when forest seedlings don't have to compete with the nutrient-absorbing fynbos roots, they grow much better. Choice B is incorrect. Although the text does indicate that the fynbos and the Afro-temperate forest are distinct from, and adjacent to, each other and that there is "almost no transition space" (area of mixed fynbos and forest plants), the experiment shows that forest plants can grow in soil occupied by fynbos roots, just not very well. So, the idea that the fynbos roots physically obstruct forest plants is unsupported by the text. Choice D is incorrect because it directly contradicts the text's claim that fynbos roots absorb nutrients from, and therefore deplete, the soil they're in, and furthermore, nothing in the text suggests that the habitat of the fynbos is harsh in general.

Q2

B

- B. It facilitates the maneuvering of particles without directly heating the particles themselves.
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Choice B is the best answer. The text says that the glass beads get the "focus" of the light beams so that the particles don't overheat. From this, we can infer that the beads allow the particles to be manipulated without being directly heated by the light beams.

Choice A is incorrect. The text never says that attaching the glass beads saves time in any way. Choice C is incorrect. The text never says that attaching the glass beads allows researchers to use weaker light beams. Choice D is incorrect. The text doesn't say that the particles can transfer heat to the glass beads—rather, it says the heat from the light focuses on the glass beads instead of the particles.

Q3

A

- A. As a dog, Karenin possesses a sense of time that involves a strong preference for predictability and an aversion to disruption.
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Choice A is the best answer because it most accurately states the main idea of the text. The text begins by stating that Tomas and Tereza's dog Karenin felt disrupted by a recent move because of his dislike of change. The text then goes on to suggest that this is because the way a dog experiences time differs from the way humans experience time: time for a dog doesn't move linearly, going "on and on, from one thing to the next," but instead moves circularly, "like the hands of a clock." That is, time for a dog is experienced as a cyclical pattern characterized by routine and predictability, with each day "following the same path." The text then concludes by providing examples of seemingly insignificant changes in routine that profoundly "disturbed [Karenin's] sense of time," causing him to feel displeasure. Thus, the main idea of the text is that Karenin's sense of time as a dog involves a strong preference for predictability and an aversion to disruption.

Choice B is incorrect. Although the text emphasizes Karenin's displeasure with the recent move to a new home, it doesn't suggest that the move has made his negative responses more pronounced than they once were. Rather, in accounting for Karenin's displeasure with the move to Switzerland, the text explains that Karenin generally has a negative response to any kind of change. Choice C is incorrect because the text doesn't suggest that Karenin comprehends time similarly to how Tomas and Tereza comprehend it. On the contrary, the text strongly implies a contrast between dogs' circular experience of time with the way humans experience time as a straightforward progression that can be "plotted on a straight line." Choice D is incorrect because the text provides no indication that a change in the places and objects surrounding Karenin causes him to feel as though time is accelerating. Although the text does use the language of "dash[ing] madly ahead" in relation to time, the phrase appears in the context of a comparison illustrating how dogs experience time: time for a dog moves just as the hands of a clock do, in a circle and "unwilling to dash madly ahead"—that is, always in a regular and predictable way.

Q4

D

- D. The reputation of Smith's metaphor of the invisible hand is not due to the importance of the metaphor in Smith's work but rather to the promotion of the metaphor by some later economists for their own ends.

Choice D is the best answer because it most accurately states the main idea of the text. The text explains that economist Adam Smith's famous metaphor of the invisible hand was putatively (that is, widely assumed but not proven) intended to illustrate a robust model (a consistently accurate generalization) of how individuals pursuing their own economic interests can create broader benefits for the population. The text then emphasizes the lack of affirmative evidence for this idea by calling out the term "putatively," and explaining that, according to Gavin Kennedy, Smith used the metaphor only once in his works, in reference to specific circumstances related to the now-outdated economic view known as mercantilism, and that the metaphor didn't garner much attention until economists in the twentieth century held it up as a paradigm (a theoretical framework in the field) and thereby implied that Smith shared some of their views on economics. By emphasizing "putatively," the text implies that there is no independent reason to believe that Smith would agree with the metaphor's use outside of the specific context for which he wrote it and that, therefore, the twentieth-century economists who used it did so to support their own views without regard for the metaphor's importance to Smith's work.

Choice A is incorrect. Although the text indicates that Smith's metaphor was largely ignored until some twentieth-century economists revived it and bolstered its status, the text suggests that the later economists used Smith's metaphor to self-servingly boost their own work while ignoring the original context in which Smith wrote it. Moreover, the statement in this choice fails to reflect the text's emphasis on Smith's limited use of the metaphor in his work. Choice B is incorrect. Although the text indicates that some twentieth-century economists altered the significance of Smith's metaphor, the text doesn't suggest that the metaphor is a "useful model" of how aggregate benefits arise from individuals' selfish actions, let alone that this usefulness is unaffected by taking the metaphor out of its original context. Choice C is incorrect. Although the text indicates that Smith's metaphor was intended as a model of how individuals acting in their own interest produce aggregate benefits and it was written within the context of the now-outdated economic theory of mercantilism, these points are subordinate to the primary idea in the text, which is that Smith's use of the metaphor was tightly constrained but twentieth-century economists ignored the original context so that they could use the metaphor to suggest, without support, that Smith would agree with their economic views.

Q5

C

- C. It should be treated as a biosignature gas if detected in the atmosphere of a rocky planet but not if detected in the atmosphere of a mini-Neptune.
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Choice C is the best answer because it states a conclusion the researchers likely agree with, given the details in the text. The text explains that a biosignature gas is a gas that can be used as an indicator that a planet harbors some form of life and some astronomers have proposed that NH_3 could serve as a biosignature gas. The researchers evaluating this claim found that the atmosphere of rocky planets would be unlikely to reach "detectably high levels" of NH_3 without biological activity, which would support the proposal of NH_3 serving as a biosignature gas. However, the text also states that mini-Neptune planets can produce NH_3 in the absence of biological activity. Thus, the text is structured to lead to the conclusion that detectable levels of NH_3 in the atmospheres of rocky planets could constitute a biosignature, but that is not the case for detectable levels of the gas in the atmospheres of mini-Neptune planets.

Choice A is incorrect because the text indicates that biological activity likely accounts for detectable levels of NH_3 in the atmospheres of rocky planets but mini-Neptune planets can have detectable levels of NH_3 in their atmospheres in the absence of biological activity. Therefore, both rocky planets and mini-Neptune planets can have detectable levels of atmospheric NH_3 . Choice B is incorrect because the text states that for NH_3 to reach detectable levels in the atmospheres of rocky planets likely means they harbor biological activity, meaning that rocky planets with detectable NH_3 usually harbor biological activity. However, that does not entail that every rocky planet with biological activity will have detectable levels of NH_3 in their atmospheres. Choice D is incorrect because the text claims only that some astronomers have proposed using NH_3 as a biosignature gas without mentioning a minimum concentration of atmospheric NH_3 that must be met for it to function as a biosignature gas.

Q6

C

- C. The cerebellum has primarily been thought to regulate motor functioning, but in recent years neuroscience researchers have been uncovering additional functions.
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Choice C is the best answer. The text says that the cerebellum has long been thought to regulate motor functioning, but new research shows that it may also have other functions—including regulating emotion and social behavior and some connection to motivation and rewards processing.

Choice A is incorrect. The VTA is described as “a center for motivation and reward processing,” and the discovery of the pathway between the VTA and the cerebellum supports the theory that the cerebellum is involved in functions other than motor coordination. Choice B is incorrect. The text says that recent research has identified this pathway, but it doesn’t discuss any previous “widely accepted claims” about the pathway’s function. The “widespread acknowledgement” mentioned early in the passage is about the cerebellum alone, not its connection to the VTA. Choice D is incorrect. The text never discusses any technological limitations or any new technologies.

Q7

D

D. It is farther north than any other nesting site discovered in South America.

Choice D is the best answer because it presents a statement about the site discovered by the researchers that is supported by the text. The text discusses Fiorelli and colleagues' discovery of egg clutches, single eggs, and eggshells in a Brazilian mine. According to the text, the presence of these eggs, which are from the Late Cretaceous period, led the researchers to conclude that the location was once a nesting and breeding site for titanosaurs. The text then explains that the finding is important because of the "previous lack of known nesting sites in northern regions of South America." If there haven't been any other discoveries of a nesting site in South America's northern regions and the site in the Brazilian mine is the first, then the text strongly suggests that the site is farther north than other nesting sites that have been discovered in South America.

Choice A is incorrect because the text doesn't suggest that the site discovered by Fiorelli and colleagues is the earliest titanosaur nesting and breeding site known to paleontologists but rather that it's the first nesting site found in northern regions of South America. Moreover, the text doesn't suggest how the timeline of the newly discovered site compares with that of other titanosaur nesting and breeding sites. Choice B is incorrect because there is no mention in the text about any difficulties that Fiorelli and colleagues faced when they were excavating the nesting and breeding site in the Brazilian mine. Choice C is incorrect because the text doesn't support the idea that the nesting and breeding site in the Brazilian mine was occupied by sauropods other than titanosaurs. The text simply mentions that titanosaurs are sauropod dinosaurs and presents the researchers' conclusion that the site they discovered was for titanosaurs.

Q8

C

- C. Miranda's ability to summon details of an experience she had before arriving on the island suggests that she may also be able to summon details of her arrival on the island.
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Choice C is the best answer because it presents a point that Prospero makes about Miranda. The text begins with Miranda responding to Prospero's claim that she probably doesn't remember her life before the island. She describes a distant memory, asking if she had "four or five women" caring for her. Prospero confirms this ("thou hadst") and then asks what else she can recall, stating that because she remembers one thing from the time before she lived on the island ("ought ere thou camest here"), she might also remember arriving on the island ("how thou camest here"). That is, Prospero indicates that Miranda's ability to summon details of an experience from before her arrival on the island suggests that she may also be able to summon details of her arrival.

Choice A is incorrect because Prospero doesn't say anything about how Miranda presents her early childhood memory of having several people care for her—he indicates only his surprise that she remembers something from so long ago ("the dark backward and abysm of time") and his thought that she may also be able to recall another childhood event: her arrival on the island. Further, Prospero doesn't suggest that he believes Miranda is discontent in her current circumstances. Choice B is incorrect. Although Miranda suggests some doubt about her recollection of a place other than the island, in that she describes it as "like a dream" she can't be certain of, Prospero doesn't mention her uncertainty—though he himself wonders how it is possible that she correctly recalls something from so long ago ("abysm of time"). Rather than indicating that doubt is making Miranda reluctant, Prospero simply states that Miranda may have the ability to remember traveling to the island. Choice D is incorrect. Although Miranda describes a vague impression of several people who cared for her as a young child and states that it is "rather like a dream" instead of something she is certain is real, Prospero doesn't indicate that Miranda is remembering a childhood daydream. Instead, he confirms that what Miranda remembers actually happened, answering her question about having had caregivers with "thou hadst," and asks her what else she can remember from long ago.

Q9

C

- C. Wild animals such as the Western scrub-jay can be effectively studied in their natural habitat and in the lab, but each approach has drawbacks that could affect the accuracy of the findings.
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Choice C is the best answer because it most accurately states the main idea of the text. The text begins by explaining that wild animals can be studied in their natural habitat or in a laboratory setting, with each setting offering unique advantages to researchers. The text then highlights an instance in which Western scrub-jays were studied in both settings but with conflicting results, indicating that one or both studies may have failed to account for the disadvantages of its research setting. Thus, the main idea of the text is that while wild animals can be effectively studied in natural or lab settings, there are drawbacks to each that need to be considered to ensure accurate results.

Choice A is incorrect because the text does not position one study setting (natural or lab) as superior to the other; rather, the text states that each one has advantages over the other. Choice B is incorrect. The text provides a hypothetical example of two studies in different environments with conflicting results, but this single example cannot be extrapolated to a general claim about the likelihood that results of studies in different environments will conflict. Additionally, the text does not assert anything about how researchers can or cannot resolve conflicting study results. Choice D is incorrect because the text does not state that discrepancies between natural-habitat and lab-based animal behavior studies are due to both of the designs being flawed. Rather, the text states that the conflict in results can be the consequence of one or both of the studies having failed to account for some factor.

Q10

C

C. Its apparent obscurity can be resolved when considered in the proper cultural context.

Choice C is the best answer because it presents a statement about the difrasismo *in cuauhtli in ocelotl* that is directly supported by the text. The text begins by describing difrasismo, a device used in Classical Nahuatl poetry. The text then mentions the device's two functions: a formal one (giving structure to lines of verse) and a ritualistic one. The text indicates that the relation between the words in a difrasismo may appear tenuous without the additional information supplied by Aztec ceremonial culture but that the meaning becomes intelligible in the context of that information. Therefore, the difrasismo's apparent obscurity can be resolved when considered in the proper cultural context.

Choice A is incorrect because the text doesn't indicate that the two nouns used in a difrasismo are semantically equivalent; instead, the text indicates that the two nouns used in a difrasismo make up a single metaphor whose meaning is often intelligible only in the context of information supplied by Aztec ceremonial culture. Choice B is incorrect because the text doesn't indicate that there's a relationship between the formal function of the difrasismo and the difrasismo's intelligibility. Additionally, the text suggests that present-day readers who are familiar with Aztec ceremonial culture wouldn't find the difrasismo to be unintelligible. Choice D is incorrect because the text doesn't indicate that the frequency of difrasismo's use in Classical Nahuatl is a necessary feature of intelligibility: the text indicates that an infrequently used difrasismo would presumably also be intelligible to members of an Aztec audience who are sufficiently familiar with Aztec ceremonial culture.

Notes

Accept equivalent forms (e.g. 0.25 $\equiv$ 1/4). For grid-in items, decimal and fraction answers are both valid.

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